John Hongyu Meng

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Working Experience & Education

Guangdong Institute of Intelligence Science and Technology *Zhuhai, China*
Principal Investigator *August 2026 - Present*
- Principal Investigator and Head of the Predictive Coding and Cognitive Systems Group

New York University *New York, USA*
Postdoctoral Researcher *October 2019 - July 2026*
- Supervisor: Dr. Xiao-Jing Wang

Northwestern University *Evanston, IL*
Applied Math PhD *September 2013 - March 2020*
- Advisor: Dr. Hermann Riecke

Peking University *Beijing, China*
BS Applied Math Major *Sep. 2009 - Jun. 2013*
- Advisor: Dr. Ruo Li

Publications

Meng JH*, Kang Y*, Lai A, Feyerabend M, Inoue W, Martinez-Trujillo J, Rudy B, Wang X-J. In Search of Transcriptomic Correlates of Neuronal Firing-Rate Adaptation across Subtypes, Regions, and Species: A Patch-seq Analysis. iScience. 2026. (Accepted) (BioRxiv version: https://doi.org/10.1101/2024.12.05.627057)

Froudist-Walsh S, Ivanov TG*, Fantacci T*, Magrou L*, Cho H, Russo E, Busch AN, Muller LE, **Meng JH**, Murray JD, Wang X-J. Mechanistic computational models of cognition across scales and species. Biological Psychiatry. 2026. (Accepted)

Meng JH, Wang X-J. Global error signal guides local optimization in mismatch calculation. Nature Communications. 2026, Mar 12;17:3868. (https://www.nature.com/articles/s41467-026-70354-x)

Meng JH, Ross JM, Hamm JP, Wang X-J. Duet model unifies diverse neuroscience experimental findings on predictive coding. (Submitted) BioRxiv. 2025. (https://doi.org/10.1101/2025.07.12.664417)

Magrou L, Froudist-Walsh S, Wang X-J, **Meng JH**, DeFelipe J, Murray JD. Principles of large-scale cortical organization across species: network properties and hierarchical gradients (Submitted)

Dellal S, Zurita H, Valero M, Abad-Perez P, Kruglikov I, **Meng JH**, Prönneke A, Hanson JL, Mir E, On-garo M, Wang X-J, Buzsáki G, Machold R, Rudy B. Inhibitory and disinhibitory VIP IN-mediated circuits in neocortex. BioRxiv. 2025. (https://doi.org/10.1101/2025.02.26.640383)

Aizenbud I, Audette N, Auksztulewicz R,... **Meng JH***,..., Xiong, Y. S. Neural mechanisms of predictive processing: a collaborative community experiment through the OpenScope program. (Submitted, authors ordered by last name) arXiv. 2025. (https://arxiv.org/abs/2504.09614)

Machold R, Dellal S, Valero M, Zurita H, Kruglikov I, **Meng JH**, Hanson JL, Hashikawa Y, Schuman B, Buzsáki G, Rudy B. Id2 GABAergic interneurons comprise a neglected fourth major group of cortical inhibitory cells. eLife. 2023 Sep 4;12:e85893. (https://elifesciences.org/articles/85893)

Meng JH, Schuman B, Rudy B, Wang X-J. Mechanisms of dominant electrophysiological features of four subtypes of layer 1 interneurons. Journal of Neuroscience. 2023 May 3;43(18):3202-18. (https://doi.org/10.1523/JNEUROSCI.1876-22.2023)

Meng JH, Riecke H. Structural spine plasticity: Learning and forgetting of odor-specific subnetworks in the olfactory bulb. PLoS Computational Biology. 2022 Oct 24;18(10):e1010338. (https://doi.org/10.1371/journal.pcbi.1010338)

Saha S, **Meng JH**, Riecke H, Agoranos G, Sailor KA, Lledo PM. Interneuron activity-structural plasticity association is driven by context-dependent sensory experience. BioRxiv. 2021 (https://doi.org/10.1101/2021.01.28.428612)

Meng JH, Riecke H. Synchronization by uncorrelated noise: interacting rhythms in interconnected oscillator networks. Scientific Reports 2018;8.1:6949. (https://www.nature.com/articles/s41598-018-24670-y)

Conference Abstracts

Meng JH, Wang X-J. Alignment between stimuli and prediction can explain deviant detection in a parsimonious model. Poster Presentation. Cosyne, Lisbon, Portugal, 2024.

Conference Presentations

Meng JH, Ross JM, Hamm JP, Wang X-J. Duet Predictive Coding Unifies Mismatch Response and Deviant Detection: Theoretical Predictions and Data Validation. (Accepted) Session Number: PSTR451. 2025 Neuroscience Meeting Planner. San Diego, CA. Society for Neuroscience, 2025.

Jin Y*, **Meng JH***, Lai A, Wang X-J. Large Sag Conductance Masked by Low Input Resistance in Parvalbumin Interneurons: A Patch-seq Analysis. Session Number: LBP022. 2025 Neuroscience Meeting Planner. San Diego, CA. Society for Neuroscience, 2025.

Meng JH*, Wang X-J. E-I pathway balance introduced by three-factor learning can explain visuomotor mismatch and deviation detection in a parsimonious model. Program No. PSTR478.09. 2024 Neuroscience Meeting Planner. Chicago, IL. Society for Neuroscience, 2024.

Dellal S, Zurita H, Valero M, Abad-Perez P, Kruglikov I, **Meng JH**, Prönneke A, Hanson JL, Mir E, On-garo M, Wang X-J, Buzsáki G, Machold R, Rudy B. Distinct inhibitory and disinhibitory circuits mediated by neocortical VIP interneuron subtypes. Program No. PSTR117.08. 2024 Neuroscience Meeting Planner. Chicago, IL. Society for Neuroscience, 2024.

Meng JH, Wang X-J. Alignment between stimuli and prediction can explain deviant detection in a parsimonious model. Poster Presentation. NeuroAI, Washington, WA, USA, 2024.

Meng* JH, KANG Y*, Lai A, RUDY B, WANG X-J. In search of Genetic Correlates of Neuronal Firing-Rate Adaptation: A Patch-seq Analysis of Transcriptomic Subtypes. Present No. PSTR525.28, Neuroscience Meeting Planner. Chicago, IL. Society for Neuroscience, 2023.

Meng JH, Schuman B, Rudy B, Wang X-J. Mechanisms of dominant electrophysiological features of four subtypes of layer 1 interneurons. Poster Presentation. CCCN, Beijing, China, 2023.

Meng JH*, Kang Y*, Rudy B, Wang X-J. Analysis of Parvalbumin and Somatostatin Interneurons Patchseq data reveals that Spike frequency Adaptation is a Subtype Identifier and Its Relationship with Fast-spiking Feature. Program No. 748.14. 2022 Neuroscience Meeting Planner. San Diego, CA. Society for Neuroscience, 2022.

Meng JH, Schuman B, Rudy B, Wang X-J. Mechanisms of dominant electrophysiological features of four subtypes of layer 1 interneurons. Abstract Presentation (Virtual Poster) P947. SfN, Chicago, IL, 2021.

Meng JH, Saha S, Sailor K, Lledo PM, Riecke H. Hebbian model of the structural plasticity in the olfactory bulb. Poster Presentation N24. SfN, Chicago, IL, 2019.

Meng JH, Riecke H. Synchronization by uncorrelated noise: interacting rhythms in interconnected neuronal networks. Poster Presentation P116. OCNS 2018, Seattle, IL, 2018.

Meng JH, Xu X and Riecke H. Independent noise synchronizing population rhythms of networks of pulse-coupled oscillators. Talk in minisymposium MS27 ”The Dynamics and Function of Neuronal Networks - Part I”. SIAM conference on the application of dynamical systems. Snowbird, USA, 2017.

Park J, **Meng JH** and Riecke H. Adaptive neuronal networks in olfaction: mechanism of network evolution and information processing. Poster Presentation. SIAM, Snowbird, USA, 2017.

Meng JH and Riecke H. Independent noise can synchronize interacting networks of pulse-coupled oscillators. Poster Presentation. Complex Network Workshop, Evanston, USA, 2015.

Karamchandani A, **Meng JH** and Riecke H. Subthreshold oscillation, multiple delays, and rhythms in the olfactory system. Poster Presentation. ICIAM 2015, Beijing, China, 2015.

Invited Talks

Duet Predictive Coding: A Unified Framework for Surprise Detection in the Sensory Cortices
- Allen Institute, online, hosted by Jerome Lecoq, Seattle, USA, Jan 2026
- Swiss Federal Institute of Technology in Lausanne, online, hosted by Gerstner lab, Lausanne, Switzerland, Nov 2025
- Nathan S. Kline Institute for Psychiatric Research, hosted by Hamm lab, NY, USA, Oct 2025
- University of Bristol, hosted by Froudist-Walsh lab, online, Bristol, England, Sep 2025
- Friedrich Miescher Institute for Biomedical Research, online, hosted by Keller lab, Basel, Switzerland, Aug 2025

A minimal model that unifies local predictive coding protocols: Mechanisms and Predictions
- Flatiron Institute, New York, NY, USA, Dec 2024

In search of Genetic Correlates of Neuronal Firing-Rate Adaptation: A Patch-seq Analysis of Transcriptomic Subtypes
- Swartz Foundation Meeting. Seattle, WA. USA, July 2024

Insights from cell-level observations to distributed cognitive functions: L1 interneuron modeling and Patchseq data analysis
- NYU-ECNU Institute of Brain and Cognitive Science (online), NYU Shanghai. Shanghai, China, Sep 2023.
- Institute of Non-Equilibrium Systems, Beijing Normal University, Beijing, China, Aug 2023.
- Research Center for Brain-inspired Intelligence, Xi'an Jiaotong University, Xi'an, China Aug 2023.
- Lingang Laboratory of Shanghai, Shanghai, China, Aug 2023.
- Research Institute of Intelligent Complex Systems, Fudan University, Shanghai, China, Aug 2023.
- Brain Cognition and Brain Disease Institute of SIAT at CAS, Shenzhen, China, Aug 2023.

Teaching Experience

NEURL-GA 3042 Computational Neuroscience *New York University, USA*
Adjunct Instructor *fall, 2023*
- Designed and led technique-oriented interactive recitations.

Computational & Cognitive Neuroscience Summer School *Suzhou, China*
Teaching Assistant *Summer, 2023*
- Developed and delivered technical tutorials.
- Guided students in designing, advancing, and refining summer school projects.

NEURL-GA 3042 Computational Neuroscience *New York University, USA*
Invited Lecturer *Fall, 2022*
- Delivered a guest lecture on firing-rate patterns and short-term plasticity.

Engineering Analysis 4, GEN_ENG 205-4 *Northwestern University, USA*
Teaching Assistant *Winter, 2018*

Engineering Analysis 4, GEN_ENG 205-4 *Northwestern University, USA*
Head Teaching Assistant *Fall, 2016*
- Managed a team of 7 TAs supporting 200 students.

Methods of Applied Mathematics, ES_APPM 311 *Northwestern University, USA*
Teaching Assistant *Fall, 2015*

Advanced Calculus, Math 234 *Northwestern University, USA*
Tutor *Spring, 2014*

Mentoring

Yufei Jin
Master's Student, New York University *2024-Present*
- Supervising a Patch-seq analysis project on H-current differences across interneuron subtypes.

Alan Lai

Ph.D. Candidate, New York University *2023-Present*

- Mentoring on building a Hodgkin—Huxley model to explain firing-rate adaptation differences across interneuron subtypes.

Yijie Kang

Master’s Student, New York University *2021-2023*

- Supervised and Collaborated on Patch-seq analyses of firing-rate adaptation across interneuron subtypes.
- Now a Graduate Student at Stony Brook University.

Academic Service

Reviewed for Journals: Communications Biology, PRX Life, Chaos, ENeuro.